

Preprocessing

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Prerequisite of doing data science

◎ The majority of the time and effort to:

- Creating
- Cleaning
- Updating

◎ Because:

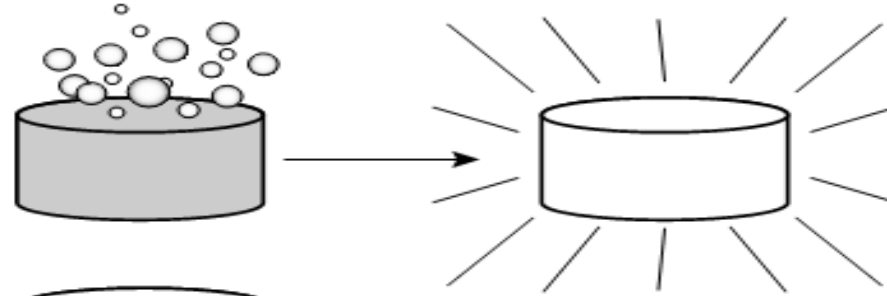
- Information from many sources
- In different formats
- Data and problems change frequently

Data quality

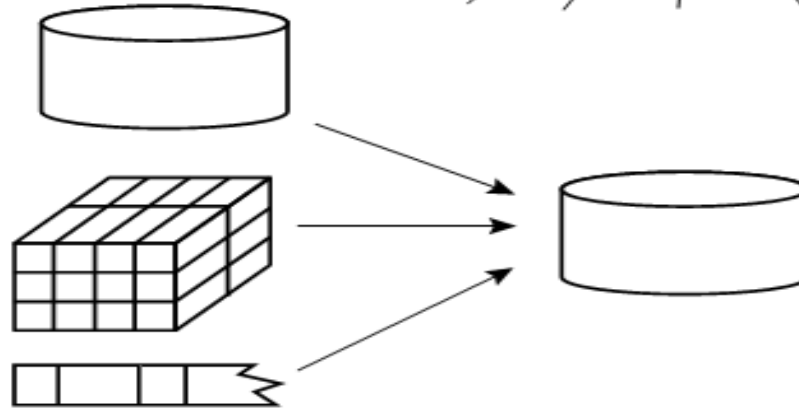
- ◎ **Missing, incomplete**: missing attribute value, missing attributes of interest, or only contains integrated data
 - Example : age, weight = " "
- ◎ **Noise**: contain errors or outliers
 - Example: salary = "-100 000"
- ◎ **Conflict**: there is inconsistency in the code or in the name
 - age =42 , birth = 03/07/1997;
 - US=USA?
 - Was rating "1, 2, 3", now rating "A, B, C"

Solutions? (1/2)

Data cleaning



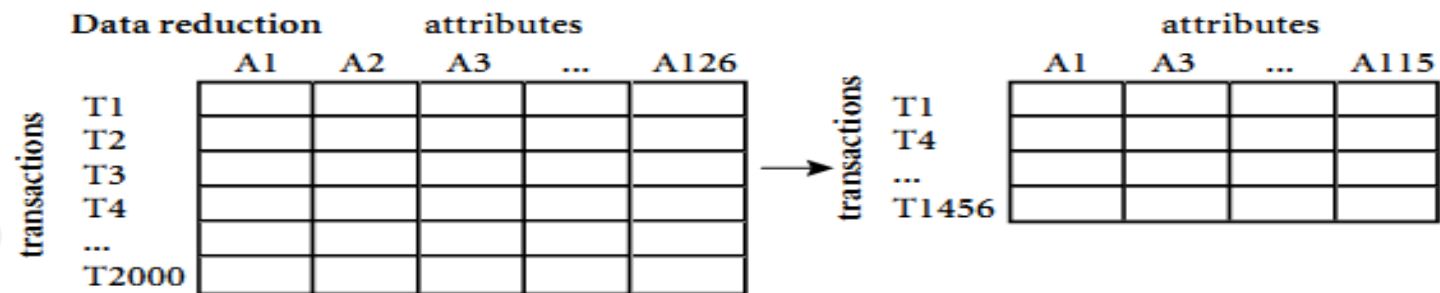
Data integration



Data transformation

$-2, 32, 100, 59, 48 \longrightarrow -0.02, 0.32, 1.00, 0.59, 0.48$

Data reduction



Filtering Out Missing Data

© Different types of missing values:

- Standard Missing Values
- Non-standard Missing Values
- Unexpected Missing Values

Filtering Out Missing Data

◎ Different types of missing values:

- Standard Missing Values
 - Blank cells, NaN, n/a

Student_Id	Math	English
1	70	60
2		55
3	45	NaN
4	75	50
5	999999	75
6	90	X
7	95	80
8	??	57
9	80	na
10	n/a	64

Highlighted all the missing values in the dataset

Student_Id	Math	English
1	70	60
2		55
3	45	NaN
4	75	50
5	999999	75
6	90	X
7	95	80
8	??	57
9	80	na
10	n/a	64

Highlighted only the standard missing values in the dataset

Filtering Out Missing Data

© Different types of missing values:

- Standard Missing Values
- Non-standard Missing Values
 - Sometimes missing values will be entered like .., __, —, missing, na, @,??,***, etc.

Student_Id	Math	English
8	??	57
9	80	na

Filtering Out Missing Data

© Different types of missing values:

- Standard Missing Values
- Non-standard Missing Values
- **Unexpected Missing Values**
 - Identified based on the context

Student_Id	Math	English
5	999999	75

Fill Missing Values

◎ Different ways to fill the missing values:

- Mean/Median, Mode
- backward fill , forward fill
- interpolate
- replace

Fill Missing Values

◎ Different ways to fill the missing values

- Mean/Median, Mode
 - Numerical Data → Mean/Median
 - Categorical Data → Mode

Fill Missing Values

◎ Different ways to fill the missing values:

- Mean/Median, Mode
- backward fill , forward fill

	Day	Temp
0	Day 1	33.0
1	Day 2	33.0
2	Day 3	35.0
3	Day 4	NaN
4	Day 5	38.0
5	Day 6	37.0
6	Day 7	39.0

Having null values

	Day	Temp
0	Day 1	33.0
1	Day 2	33.0
2	Day 3	35.0
3	Day 4	38.0
4	Day 5	38.0
5	Day 6	37.0
6	Day 7	39.0

bfill

	Day	Temp
0	Day 1	33.0
1	Day 2	33.0
2	Day 3	35.0
3	Day 4	35.0
4	Day 5	38.0
5	Day 6	37.0
6	Day 7	39.0

ffill

Fill Missing Values

© Different ways to fill the missing values:

- Mean/Median, Mode
- backward fill , forward fill
- **interpolate**

	Day	Temp
0	Day 1	34.0
1	Day 2	33.0
2	Day 3	35.0
3	Day 4	NaN
4	Day 5	NaN
5	Day 6	NaN
6	Day 7	38.0
7	Day 8	37.0
8	Day 9	39.0

Having null values

	Day	Temp
0	Day 1	34.00
1	Day 2	33.00
2	Day 3	35.00
3	Day 4	35.75
4	Day 5	36.50
5	Day 6	37.25
6	Day 7	38.00
7	Day 8	37.00
8	Day 9	39.00

interpolate

Fill Missing Values

◎ Different ways to fill the missing values:

- Mean/Median, Mode
- backward fill , forward fill
- interpolate
- **replace**

Drop Missing Values

◎ Removing the missing values in the dataset

	Student_ID	Math	English
0	1	85.0	50.0
1	2	76.0	65.0
2	3	80.0	NaN
3	4	NaN	55.0
4	5	70.0	60.0

dataset



	Student_ID	Math	English
0	1	85.0	50.0
1	2	76.0	65.0
4	5	70.0	60.0

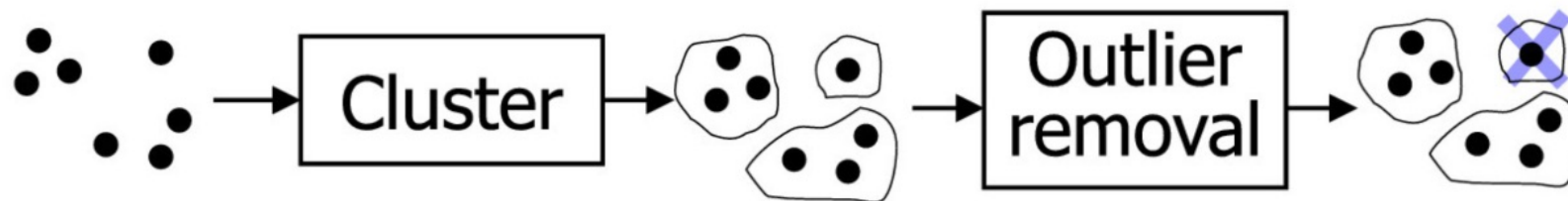
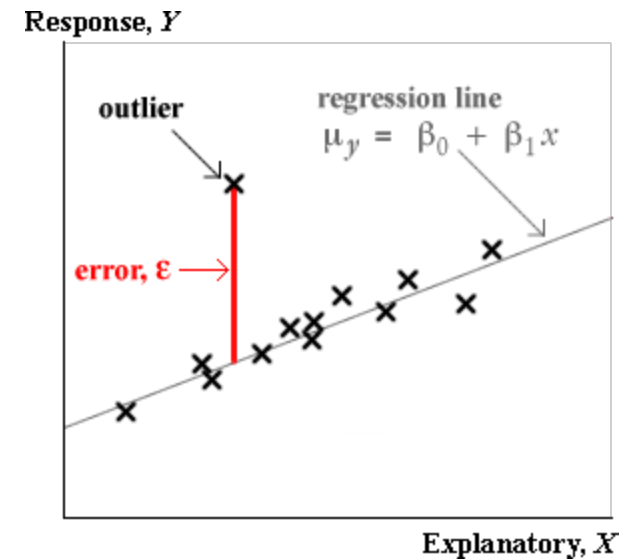
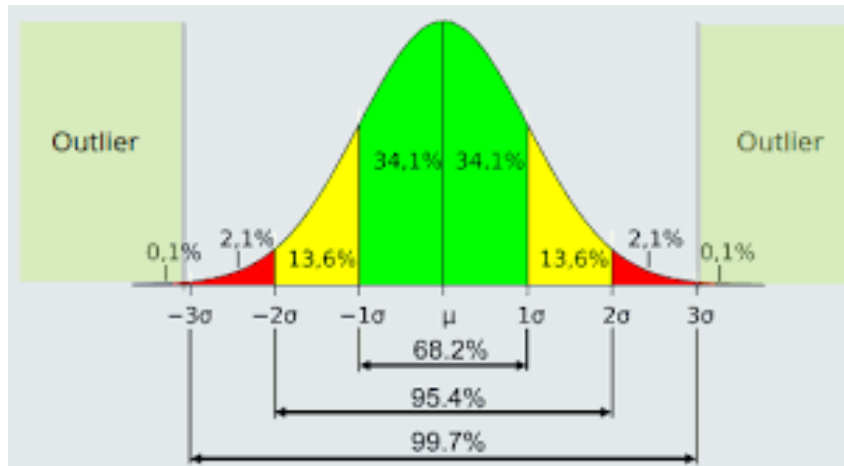
Removing all
the null values

Removing Duplicates

	A	B	C	D		A	B	C	D
1	Last Name	Sales	Country	Quarter	1	Last Name	Sales	Country	Quarter
2	Smith	\$16,753.00	UK	Qtr 3	2	Smith	\$16,753.00	UK	Qtr 3
3	Johnson	\$14,808.00	USA	Qtr 4	3	Johnson	\$14,808.00	USA	Qtr 4
4	Williams	\$10,644.00	UK	Qtr 2	4	Williams	\$10,644.00	UK	Qtr 2
5	Jones	\$1,390.00	USA	Qtr 3	5	Jones	\$1,390.00	USA	Qtr 3
6	Brown	\$4,865.00	USA	Qtr 4	6	Brown	\$4,865.00	USA	Qtr 4
7	Smith	\$16,753.00	UK	Qtr 3	7	Williams	\$12,438.00	UK	Qtr 1
8	Williams	\$12,438.00	UK	Qtr 1	8	Johnson	\$9,339.00	UK	Qtr 2
9	Johnson	\$9,339.00	UK	Qtr 2	9	Smith	\$18,919.00	USA	Qtr 3
10	Smith	\$18,919.00	USA	Qtr 3	10	Jones	\$9,213.00	USA	Qtr 4
11	Jones	\$9,213.00	USA	Qtr 4	11	Jones	\$7,433.00	UK	Qtr 1
12	Jones	\$7,433.00	UK	Qtr 1	12				
13	Smith	\$16,753.00	UK	Qtr 3	13				
14					14				
15					15				

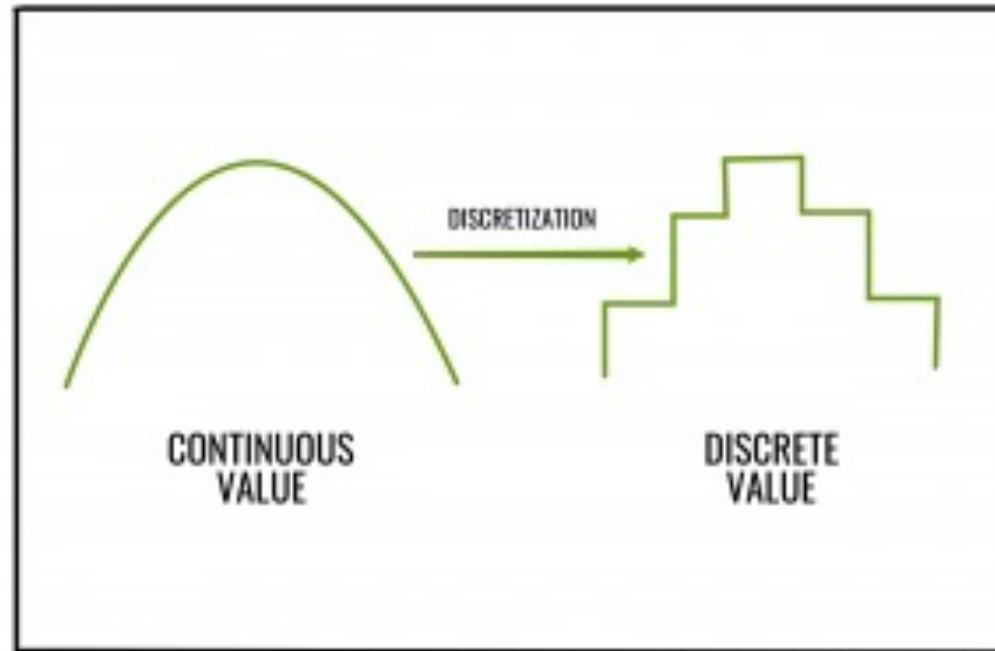
Detecting and Filtering Outliers

◎ Outliers are values higher/lower than $1.5 \times \text{IQR}$



Discretization

- © **Discretization** (quantization, binning) divides a continuous feature into a pre-specified number of categories (bins), and thus makes the data discrete.



Normalization

◎ Scale data within a small, specified range

- min-max normalization

$$v' = (v - \text{min}) / (\text{max} - \text{min})$$

- Z-score normalization

$$v' = (v - \text{mean}) / \text{stand_dev}$$

- Normalization by decimal scaling

$$v' = v / 10^j$$

where j is the smallest integer such that $\text{Max}(|v'|) < 1$

Other preprocessings

◎ Text normalization:

- converting all letters to lower or upper case
- converting numbers into words or removing numbers
- removing punctuations, accent marks and other diacritics
- removing white spaces
- expanding abbreviations
- removing stop words, sparse terms, and particular words
- text canonicalization
- stemming

Transformation into vectors

© All the ML models need the input data to be in the form of vectors.

- **String**: tf-idf, word2vec, ...
- **Images**: pixels, ...
- **Sound**: digital signals, ...
- **Categories**: label encoding, one-hot-encoding, ...



The End